Heart Failure with Reduced Ejection Fraction (HFrEF): Guideline Summary 2026

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Summary. This synthetic summary condenses the current standard of care for adults with heart failure with reduced ejection fraction (HFrEF): diagnosis, guideline-directed medical therapy (GDMT), device therapy, advanced care, emerging therapies and quality measures. It was written to demonstrate document extraction and is not clinical guidance.

1. Definition and Epidemiology

HFrEF is heart failure with a left ventricular ejection fraction (LVEF) of 40% or lower. It accounts for roughly half of heart failure cases in this summary's reference population. Patients with an LVEF between 41% and 49% are classified as mildly reduced ejection fraction and are managed separately.

Five-year mortality after a first heart failure hospitalization remains close to 50% in the reference registry, and 30-day readmission after discharge averages 21%, which makes HFrEF a priority for value-based care programs.

2. Diagnosis

Diagnosis combines symptoms, natriuretic peptides and imaging. An NT-proBNP above 125 pg/mL in the outpatient setting supports the diagnosis; transthoracic echocardiography confirms the LVEF and identifies structural causes.

Initial work-up includes ECG, renal function, potassium, iron studies and screening for ischemic heart disease. Iron deficiency (ferritin below 100 ng/mL, or 100-299 ng/mL with transferrin saturation below 20%) should be corrected.

3. Guideline-Directed Medical Therapy (GDMT)

The foundation of HFrEF care is four drug classes started together rather than sequentially: an angiotensin receptor-neprilysin inhibitor (ARNI), an evidence-based beta-blocker, a mineralocorticoid receptor antagonist (MRA) and an SGLT2 inhibitor. Each class independently reduces mortality or hospitalization.

Rapid sequencing - all four pillars within six weeks and dose titration every two weeks - is preferred over waiting for target doses of one drug before adding the next. Table 1 lists recommended agents and target doses.

Table 1. Recommended GDMT for HFrEF

Drug class	Example agents	Target dose (example)	Recommendation	Evidence
ARNI	sacubitril/valsartan	97/103 mg twice daily	Class 1; preferred over ACEi/ARB in NYHA II-III	A
ACE inhibitor	enalapril	10-20 mg twice daily	Class 1 when ARNI is not feasible	A
ARB	valsartan	160 mg twice daily	Class 1 if ACE inhibitor intolerant	A
Beta-blocker	carvedilol, metoprolol succinate, bisoprolol	carvedilol 25 mg twice daily	Class 1 for all patients with HFrEF	A
MRA	spironolactone, eplerenone	spironolactone 25-50 mg daily	Class 1 if eGFR above 30 and potassium below 5.0	A
SGLT2 inhibitor	dapagliflozin, empagliflozin	10 mg daily	Class 1 with or without diabetes	A
Loop diuretic	furosemide, torsemide	titrate to euvolemia	Class 1 for congestion	B
Hydralazine + isosorbide dinitrate	fixed-dose combination	75/40 mg three times daily	Class 1 for self-identified Black patients, NYHA III-IV	A
Ivabradine	ivabradine	7.5 mg twice daily	Class 2a; sinus rhythm with heart rate 70 or higher	B

Table 1 (continued)

Drug class	Example agents	Target dose (example)	Recommendation	Evidence
sGC stimulator	vericiguat	10 mg daily	Class 2b after a worsening HF event	B
IV iron	ferric carboxymaltose	weight-based dosing	Class 2a for iron deficiency	B

4. Device Therapy and Advanced Care

Device decisions follow at least three months of optimized GDMT, because LVEF often improves with therapy. Figure 1 summarizes the treatment algorithm, including the thresholds for an implantable cardioverter-defibrillator and cardiac resynchronization therapy.

Patients who remain in NYHA class IV despite therapy (Stage D) should be referred early to an advanced heart failure program. Palliative care is integrated alongside life-prolonging treatment, not only at the end of life.

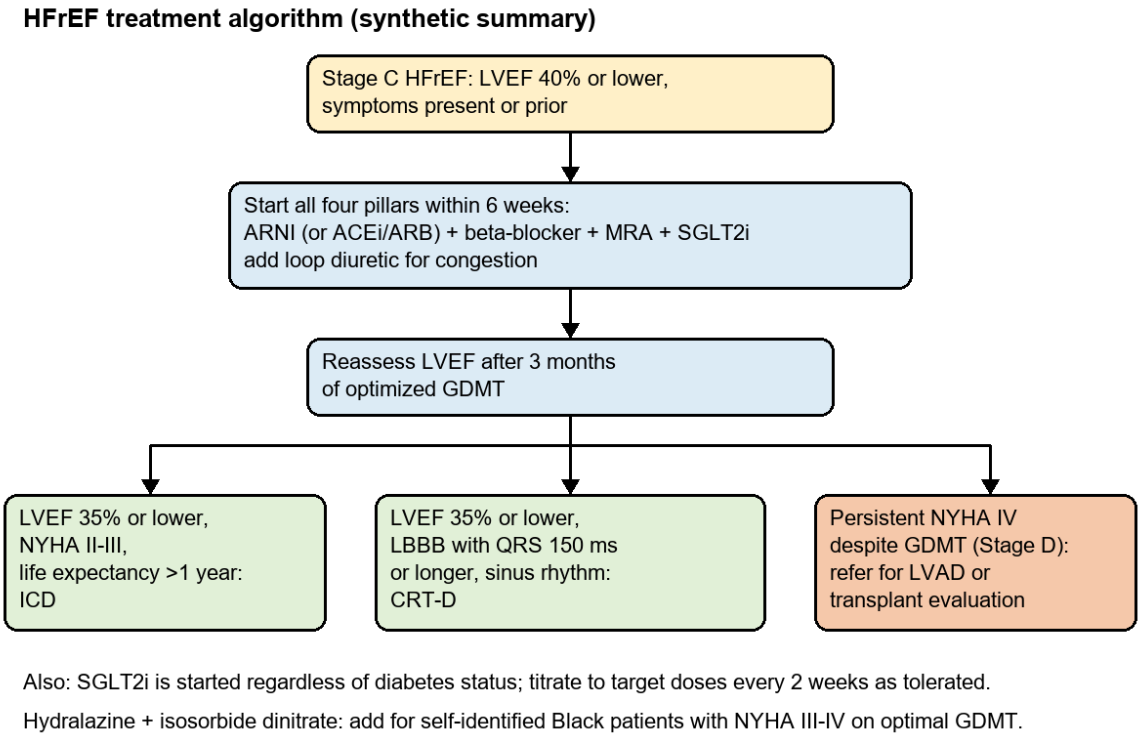


Figure 1. HFrEF treatment algorithm: GDMT initiation, reassessment and device or advanced-therapy decisions.

5. Emerging Therapies

Development is concentrated in four areas: cardiac myosin activation to improve contractility, nonsteroidal mineralocorticoid receptor antagonists with lower hyperkalemia risk, gene therapy targeting calcium handling (SERCA2a), and device-based monitoring that detects congestion before symptoms.

Digital and care-delivery programs that close the GDMT gap are a fast-growing category, because fewer than one in four eligible patients receives all four pillars at target dose.

Figure 2 shows the active synthetic pipeline by modality and the share in late-stage development. Trial-level detail (sponsors, lead sites, endpoints and timelines) is maintained in the companion registry file.

Strategy teams should track readouts expected in 2027-2029 for the late-stage programs, which could change first-line combinations and device referral volumes.

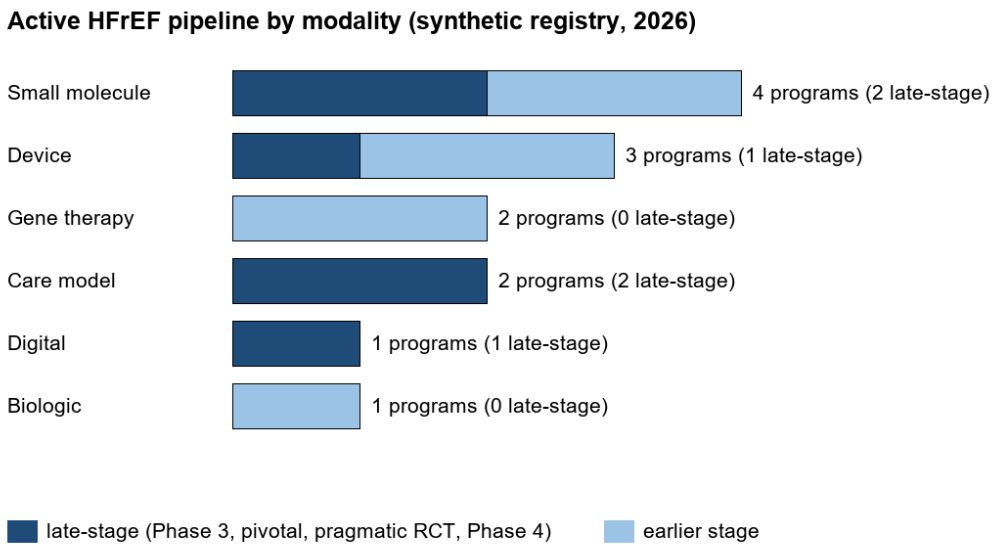


Figure 2. Active HFrEF pipeline by modality, synthetic registry 2026.

6. Quality Measures and Care Delivery

Health systems are benchmarked on the measures below. Targets are synthetic values for demonstration.

Table 2. HFrEF quality measures

Measure	Definition	Benchmark target
GDMT four-pillar rate	Eligible patients on ARNI/ACEi/ARB, beta-blocker, MRA and SGLT2i	80% or higher
7-day follow-up	Clinic or virtual visit within 7 days of discharge	60% or higher
30-day readmission	All-cause readmission within 30 days of an HF discharge	below 20%
Cardiac rehabilitation referral	Referral at discharge for eligible patients	70% or higher
Device evaluation	ICD/CRT assessment documented after 3 months of GDMT	90% or higher

a. Doses are examples for orientation only; prescribing follows the full guideline and product labeling.
b. Pipeline counts exclude completed studies. Synthetic registry snapshot dated 2026-09-30.

7. Sources

This document is a synthetic composite prepared for a portfolio demonstration. It paraphrases widely published standard-of-care principles; all figures, targets and registry data are fictional.